



VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

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NBA Accreditation for B.Tech., CE,EEE,ME,ECE,CSE,EIE,IT,AME, M.Tech,STRE, PE, AMS, SWE Programmes
Approved by AICTE, New Delhi, Affiliated to JNTUH, NIRF (2021) 135th Rank in Engineering Category
College with Potential for Excellence by UGC, JNTUH-Recognized Research Centres:CE,EEE,ME,ECE,CSE
Vignana Jyothi Nagar, Pragathi Nagar, Nizampet (S.O.), Hyderabad – 500 090, TS, India.
Telephone No:040-2304 2755/59/60, Fax:040-23042761
E-mail: postbox@vnvji.ac.in, Website: www.vnvji.ac.in



REAL TIME TRAFFIC MONITORING IN METROPOLITAN CITIES

Abstract – Batch 2

Metropolitan cities face persistent challenges in managing urban traffic, including congestion, frequent accidents, delayed public transport, and deteriorating air quality. This project presents a comprehensive data engineering solution that builds a real-time pipeline to collect, process, and analyse traffic-related data from various sources such as public APIs, IoT sensor simulations, GTFS transport feeds, and government datasets. Using tools like Apache Kafka, Spark, Airflow, and PostgreSQL/MongoDB, the system performs scalable ETL operations to clean, transform, and store large volumes of structured and unstructured traffic data.

The pipeline generates actionable insights including congestion heatmaps, accident hotspot detection, traffic violation patterns, bus delay analytics, and correlations between pollution levels and traffic density. These insights are visualized using dashboards and map-based interfaces for easy interpretation. The project aims to empower urban planners, traffic control authorities, and civic bodies with data-driven tools to make informed decisions for improving traffic flow, road safety, and environmental conditions across metropolitan cities.

Team Details

S.No	NAME	ROLL NO
1	N. AISHWARYA	23071A6901
2	G. SRIHITHA	23071A6918
3	J. DILIP REDDY	23071A6922
4	P. DAIWIK	23071A6945



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INTRODUCTION – Batch 2

Metropolitan cities face growing challenges in managing urban traffic due to increasing vehicle density, rapid urbanization, and outdated monitoring systems. Issues such as traffic congestion, frequent accidents, delayed public transport, and declining air quality significantly impact city life. This project proposes a real-time data engineering solution that collects and processes traffic-related data from diverse sources like public APIs, IoT sensor simulations, GTFS feeds, and government datasets. Using tools such as Apache Kafka, Apache Spark, Apache Airflow, and PostgreSQL/MongoDB, the pipeline performs scalable ETL operations to clean, transform, and store both structured and unstructured data.

It generates actionable insights including congestion heatmaps, accident hotspots, traffic violation trends, bus delay patterns, and pollution-traffic correlations. These insights are visualized using interactive dashboards and geospatial maps to support quick interpretation. The goal is to empower urban planners, traffic authorities, and civic bodies with data-driven tools to enhance traffic flow, road safety, and environmental sustainability in metropolitan areas.

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